LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Quick Revision: Motion

Key terms

Term	Meaning	SI unit
Distance	Total length of the path covered (scalar)	m
Displacement	Shortest distance from start to finish, with direction (vector)	m
Speed	Distance covered per unit time	m/s
Velocity	Speed in a given direction (displacement per unit time)	m/s
Acceleration	Change in velocity per unit time	m/s^2
Uniform motion	Equal distances in equal intervals of time	–

Equations of motion (uniform acceleration): $v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 - u^2 = 2as$
Average speed = total distance ÷ total time · Average velocity = (u + v)/2 (uniform acceleration only)
km/h → m/s: divide by 3.6 · Uniform circular motion: $v = 2\pi r/t$

Must-know points

- Displacement can be zero even when distance is not (e.g. one full round of a track).
- Distance–time graph: slope = speed. A straight sloping line means uniform speed.
- Velocity–time graph: slope = acceleration; **area under the graph = displacement**.
- A horizontal line on a velocity–time graph means uniform velocity (zero acceleration).
- Uniform circular motion is **accelerated** motion: the speed is constant but the direction keeps changing.
- Retardation (slowing down) is negative acceleration.

Common mistakes

- Mixing km/h with m/s in the same formula. Convert first.
- Forgetting the minus sign for retardation, or for a body thrown upwards ($a = -g$).

Quick check

1. A car speeds up from 18 km/h to 36 km/h in 5 s. Find its acceleration.

2. What does the area under a velocity–time graph tell you?